
Laminar Premixed Flames

Outline *Until now, we have covered the theory behind combustion. Specifically, we read about **thermo-chemistry** which governs the heat released and equilibrium concentrations of product species, **transport processes** which talked about the transfer of species and energy produced, **chemical kinetics** which dealt with the dynamic behavior which may or may not reach equilibrium conditions, and finally **combustion chemistry** focusing on the different chemical mechanisms and pathways that the reactant take to convert to these products. Applying these to practical scenarios is what the next part of the course is going to be concerned with. In doing so, we will divide our applications based on certain characteristics which are quite relevant in these scenarios. These criteria are the flow laminarity/turbulent behavior and if the reactants are premixed or diffusion controlled. The first module in this context talks about laminar premixed flames i.e. combustion scenarios where the reactants are premixed and the flow characteristics are laminar. This will be concerned topic for this handout.*

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1 Introduction

A premixed flame is self-sustaining propagation of a localized combustion zone at subsonic/supersonic velocities. We use the term *deflagration* in gas-dynamics to define a premixed flame travelling at subsonic velocities and *detonation* for supersonic combustion. Consider a premixed flammable mixture in a long tube, open at both ends, ignited from one end. A combustion wave will travel down the tube starting from the ignition point. A flame is caused by a self-propagating exothermic reaction which is accompanied by a reaction zone. It will propagate through a stationary gas mixture at a characteristic velocity (burning velocity). For most hydrocarbon-air stoichiometric mixtures, this velocity is about 0.4 to 0.6 m/s. For hydrogen-air mixtures, this velocity is several meters per second. The velocity of this wave is controlled by the diffusion of heat and active radicals, For a flame burning in a mixture of gases of known pressure and composition, two characteristic properties may be defined and measured, the burning velocity and the flame temperature. Flame temperature can be predicted from thermodynamic data, if we invoke the assumption of chemical equilibrium. Various flame theories attempt to predict the laminar flame propagation from physical and chemical properties; however, a closed form solution which is universal and accurate has not been possible. Historically, there have been two approaches to formulating the laminar flame propagation in premixed gases:

- Thermal propagation: the mixture is heated by conduction to the point where the rate of reaction is sufficiently rapid to become self-propagating.
- Diffusional propagation: diffusion of active species, such as atoms and radicals, from the reaction zone or the burned gas into the unreacted mixture causes reaction to occur.

In reality, diffusion of heat and active radicals happens together. The reaction zone is shown below:

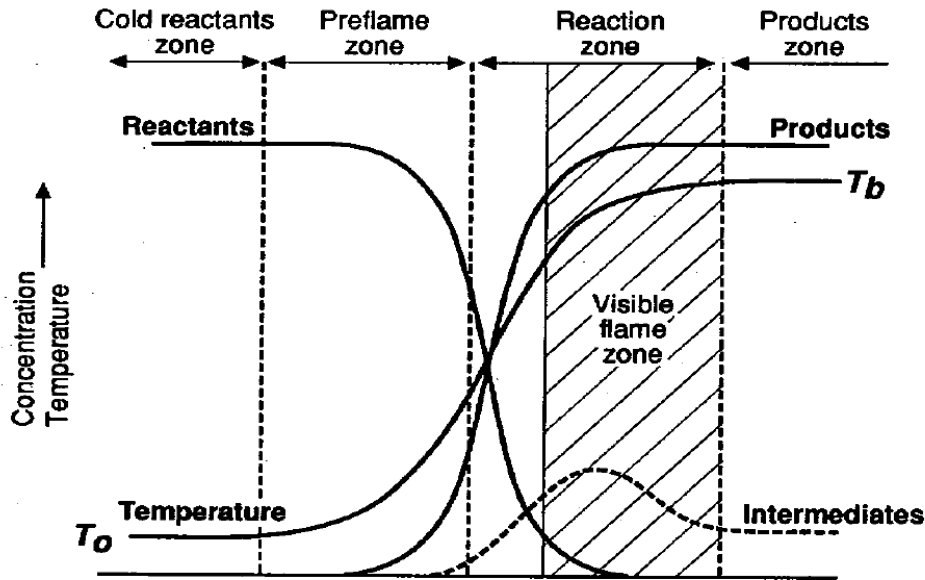


Figure 1: Schematic of a premixed flame structure for a stagnant reactant mixture.

1.1 Simplified Analysis

The objective is to find a simple analytical expression for the laminar flame speed. Assumptions are listed below:

- 1-D, constant area, steady-flow.
- Kinetic/potential energies, viscous effects, thermal radiation are all neglected.
- Pressure is assumed constant across the flame.
- Diffusion of heat and mass are governed by Fourier's and Fick's laws, respectively.
- The Lewis number is unity,

$$\text{Le} \equiv \frac{\alpha}{\mathcal{D}} = \frac{k}{\rho c_p \mathcal{D}} \quad (1)$$

which means $k/c_p = \rho \mathcal{D}$ that simplifies the energy equation.

- Individual c_p values are all equal and constant.
- Single-step exothermic reaction describes combustion.
- $\Phi \leq 1$ so that the fuel is completely consumed.

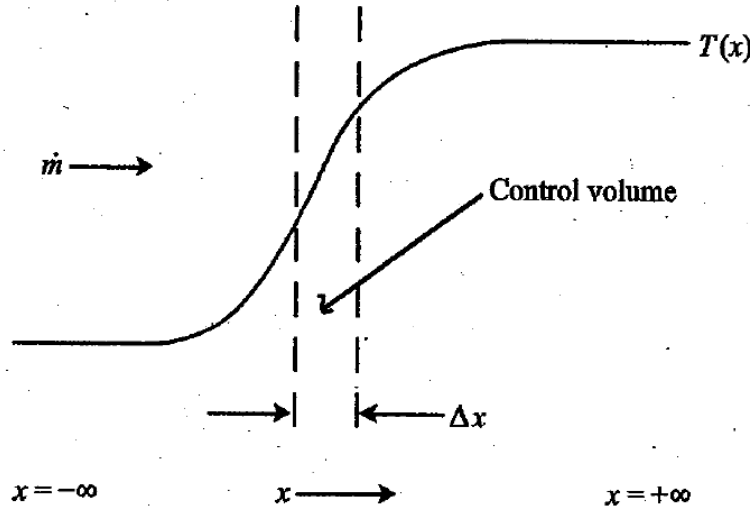


Figure 2: Reaction zone for simplified analysis

Conservation Laws:

- Mass conservation:

$$\frac{d(\rho v_x)}{dx} = 0 \quad (2a)$$

$$\dot{m}'' = \rho v_x = \text{constant} \quad (2b)$$

- Species conservation:

$$\frac{d\dot{m}_i''}{dx} = \dot{m}_i''' \quad (3)$$

With the application of Fick's diffusion law,

$$\frac{d\left[\dot{m}''Y_i - \rho\mathcal{D}\frac{dY_i}{dx}\right]}{dx} = \dot{m}_i''' \quad (4)$$

For single-step reaction,

$$1 \text{ kg fuel} + \nu \text{ kg oxidizer} \rightarrow (\nu + 1) \text{ kg products} \quad (5)$$

$$\dot{m}_F''' = \frac{1}{\nu}\dot{m}_{Ox}''' = -\frac{1}{\nu+1}\dot{m}_{Pr}''' \quad (6)$$

We can write (4) for each species:

– Fuel:

$$\dot{m}''\frac{dY_F}{dx} - \frac{d\left(\rho\mathcal{D}\frac{dY_F}{dx}\right)}{dx} = \dot{m}_F''' \quad (7a)$$

– Oxidizer:

$$\dot{m}''\frac{dY_{Ox}}{dx} - \frac{d\left(\rho\mathcal{D}\frac{dY_{Ox}}{dx}\right)}{dx} = \nu\dot{m}_F''' \quad (7b)$$

– Products:

$$\dot{m}''\frac{dY_{Pr}}{dx} - \frac{d\left(\rho\mathcal{D}\frac{dY_{Pr}}{dx}\right)}{dx} = -(\nu+1)\dot{m}_F''' \quad (7c)$$

- Energy Conservation:

$$\dot{m}''c_p\frac{dT}{dx} - \frac{d}{dx}\left[\left(\rho\mathcal{D}c_p\right)\frac{dT}{dx}\right] = -\sum h_{f,i}^o\dot{m}_i''' \quad (8a)$$

RHS of (8a) can be written as,

$$-\sum h_{f,i}^o\dot{m}_i''' = -[h_{f,F}^o\dot{m}_F''' + h_{f,Ox}^o\nu\dot{m}_F''' - h_{f,Pr}^o(\nu+1)\dot{m}_F'''] \quad (8b)$$

or

$$-\sum h_{f,i}^o\dot{m}_i''' = -\dot{m}_F'''\Delta h_c \quad (8c)$$

where Δh_c is the heat of combustion,

$$\Delta h_c \equiv h_{f,F}^o + \nu h_{f,Ox}^o - (\nu+1)h_{f,Pr}^o \quad (8d)$$

If we set $\rho\mathcal{D}c_p = k$, then (8a) becomes,

$$\dot{m}''\frac{dT}{dx} - \frac{1}{c_p}\frac{d\left(k\frac{dT}{dx}\right)}{dx} = -\frac{\dot{m}_F'''\Delta h_c}{c_p} \quad (8e)$$

The objective here is to find an expression for the laminar flame speed, which is related to the mass flux, $\dot{m}''$, by,

$$\dot{m}'' = \rho_u S_L \quad (9)$$

The approach is to assume a temperature profile that satisfies the boundary conditions, then integrate (8e).

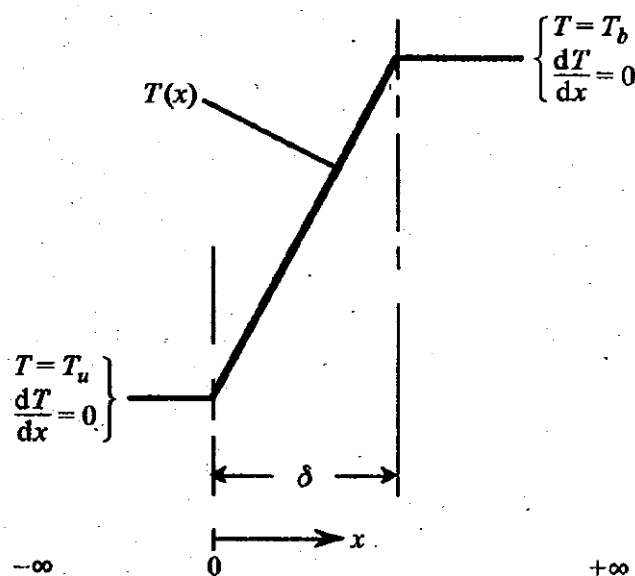


Figure 3: Assumed temperature profile and boundary condition

Boundary Conditions:

- Upstream:

$$T(x \rightarrow -\infty) = T_u \quad (10a)$$

$$\frac{dT}{dx}(x \rightarrow -\infty) = 0 \quad (10b)$$

- Downstream:

$$T(x \rightarrow +\infty) = T_b \quad (10c)$$

$$\frac{dT}{dx}(x \rightarrow +\infty) = 0 \quad (10d)$$

We assume a simple linear temperature profile that extends from T_u to T_b over the distance of δ .

Integrating (8e) over x , subject to boundary conditions above,

$$\dot{m}''[T]_{T_u}^{T_b} - \frac{k}{c_p} \left[\frac{dT}{dx} \right]_0^0 = -\frac{\Delta h_c}{c_p} \int_{-\infty}^{\infty} \dot{m}_F''' dx \quad (11)$$

$$\dot{m}''(T_b - T_u) = -\frac{\Delta h_c}{c_p} \int_{-\infty}^{\infty} \dot{m}_F''' dx \quad (12)$$

Limits of reaction rate integral can be switched to temperature from space coordinate, since $\dot{m}_F'''$ is nonzero between T_u and T_b over the interval δ ,

$$\frac{dT}{dx} = \frac{T_b - T_u}{\delta} \quad \text{or} \quad dx = \frac{\delta}{T_b - T_u} dT \quad (13)$$

then (12) becomes,

$$\dot{m}''(T_b - T_u) = -\frac{\Delta h_c}{c_p} \frac{\delta}{(T_b - T_u)} \int_{T_u}^{T_b} \dot{m}_F''' dT \quad (14)$$

If we define the average reaction rate as,

$$\bar{\dot{m}}_F''' \equiv \frac{1}{(T_b - T_u)} \int_{T_u}^{T_b} \dot{m}_F''' dT \quad (15)$$

We obtain,

$$\dot{m}''(T_b - T_u) = -\frac{\Delta h_c}{c_p} \delta \bar{\dot{m}}_F''' \quad (16)$$

In (16), we have two unknowns, $\dot{m}''$ and δ ; we need a second equation to complete the solution. If we assume that the reaction rate is much smaller within the first half of δ , i.e., between $x = -\infty$ and $x = \delta/2$, we can reevaluate (11) from $x = -\infty$ to $x = \delta/2$. Noting that at $x = \delta/2$,

$$T = \frac{T_b + T_u}{2} \quad (17)$$

and from (13) and (11), we get

$$\dot{m}'' \delta/2 - k/c_p = 0 \quad (18)$$

Simultaneous solution of (16) and (18) gives,

$$\dot{m}'' = \left[2 \frac{k}{c_p^2} \frac{(-\Delta h_c)}{(T_b - T_u)} \bar{\dot{m}}_F''' \right]^{1/2} \quad (19)$$

and

$$\delta = 2k/(c_p \dot{m}'') \quad (20)$$

Applying the definitions $S_L \equiv \dot{m}''/\rho_u$, $\alpha \equiv k/(\rho_u c_p)$, and recognizing that $\Delta h_c = (\nu + 1)c_p(T_b - T_u)$, we get,

$$S_L = \left[-2\alpha(\nu + 1) \frac{\bar{\dot{m}}_F'''}{\rho_u} \right]^{1/2} \quad (21)$$

$$\delta = \left[\frac{-2\rho_u\alpha}{(\nu + 1)\bar{\dot{m}}_F'''} \right]^{1/2} \quad (22a)$$

$$\delta = 2\alpha/S_L \quad (22b)$$

1.2 Factors Influencing Flame Velocity and Thickness

From (19) and (20), we can infer temperature dependencies of S_L and δ . First we consider the following approximate temperature scalings,

$$\alpha \propto T_u \bar{T}^{0.75} P^{-1} \quad (23)$$

$$\bar{\dot{m}}_F'''/\rho_u \propto T_u T_b^{-n} P^{n-1} \exp[-E_A/(R_u T_b)] \quad (24)$$

where n is the overall reaction order, and $\bar{T} \equiv 0.5(T_b + T_u)$. Combining the above scalings yields,

$$S_L \propto \bar{T}^{0.375} T_u T_b^{-\frac{n}{2}} \exp\left[\frac{-E_A}{2R_u T_b}\right] P^{\frac{n-2}{2}} \quad (25)$$

$$\delta \propto \bar{T}^{0.375} T_b^{\frac{n}{2}} \exp\left[\frac{E_A}{2R_u T_b}\right] P^{-\frac{n}{2}} \quad (26)$$

Strong dependence on T , i.e on both T_u and T_b . Flame thickness is inversely proportional to S_L . For $n \approx 2$, S_L is independent of P !

Empirical correlation for stoichiometric methane-air mixtures:

- Temperature:

$$S_L(\text{cm/s}) = 10 + 3.71 \cdot 10^{-4} [T_u(\text{K})]^2 \quad (27)$$

- Pressure:

$$S_L(\text{cm/s}) = 43 \cdot [P(\text{atm})]^{-0.5} \quad (28)$$

Our simplified analysis captures the effect of T , but not the effect of P .

1.3 Flame Speed Correlations

$$S_L = S_{L,\text{ref}} \left(\frac{T_u}{T_{u,\text{ref}}} \right)^\gamma \left(\frac{P}{P_{\text{ref}}} \right)^\beta (1 - 2.1Y_{\text{dil}}) \quad (29)$$

for $T_u \geq 350$ K. $T_{u,\text{ref}} = 298$ K, and $P_{\text{ref}} = 1$ atm where

$$S_{L,\text{ref}} = B_M + B_2(\Phi - \Phi_M)^2 \quad (30a)$$

$$\gamma = 2.18 - 0.8(\Phi - 1) \quad (30b)$$

$$\beta = -0.16 + 0.22(\Phi - 1) \quad (30c)$$

The values of B_M, B_2, Φ_M for a few selected fuels is given in 1.

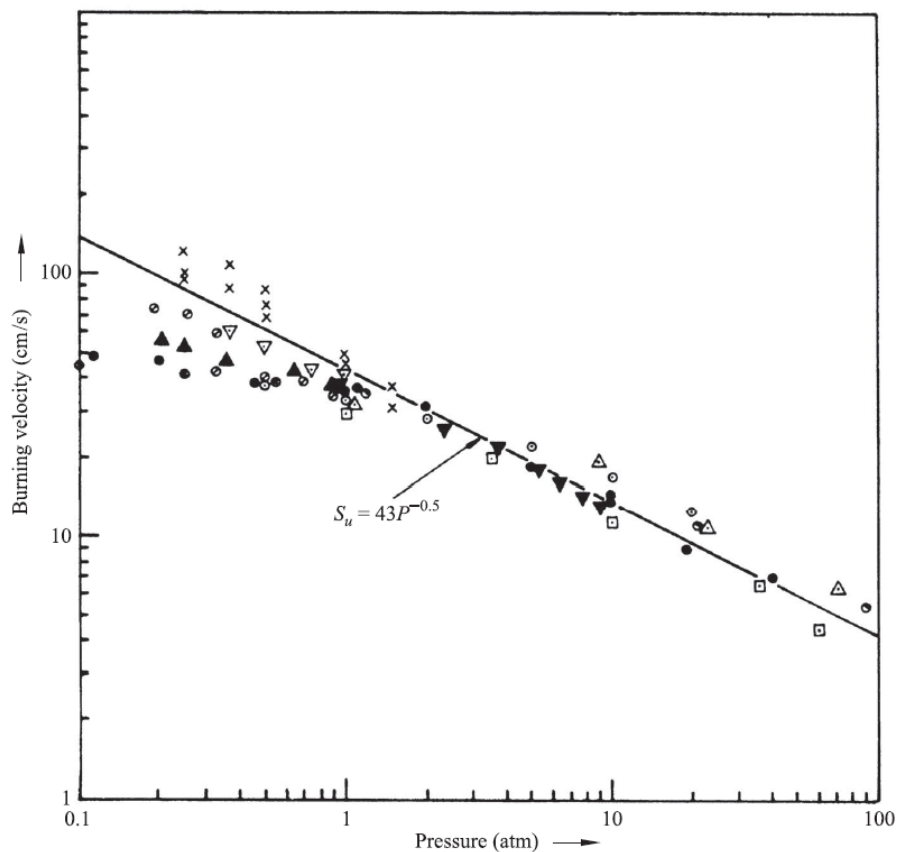


Figure 4: Effect of pressure on laminar flame speeds of stoichiometric methane–air mixtures for $T_u = 16\text{--}27^\circ\text{C}$.

Table 1: Values of B_M, B_2, Φ_M used in (30a)

Fuel	Φ_M	$B_M, \text{cm/s}$	$B_2, \text{cm/s}$
Methanol	1.11	36.92	-140.51
Propane	1.08	34.22	-138.65
Isooctane	1.13	26.32	-84.72
RMFD-303	1.13	27.58	-78.34

2 Quenching, Flammability and Ignition

So far in this chapter we have considered only the steady propagation of premixed laminar flames. We now turn our attention to what are essentially transient processes: flame quenching and ignition. Although the processes are transient, we will confine our attention to examining limit behavior, i.e., conditions under which a flame will either extinguish or not, or ignite or not, and ignore the time-dependent details of the extinction and ignition processes.

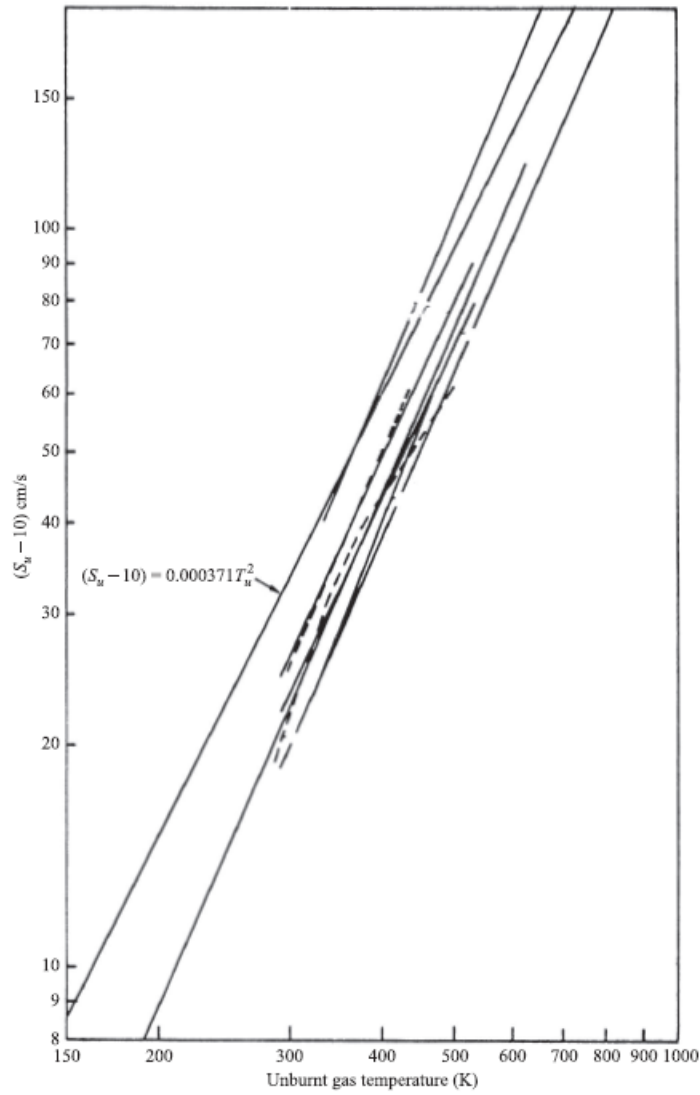


Figure 5: Effect of unburned gas temperature on laminar flame speeds of stoichiometric methane–air mixtures at 1 atm. Various lines are data from various investigators.

2.1 Quenching

A flame can be extinguished by:

- thermal effects (heat loss)
- chemical suppression
- aerodynamic effect

There are many ways in which a flame can be extinguished. For example, flames will extinguish while passing through narrow passageways. If the passageway is large enough flame will propagate through it. **Quenching distance** is the critical diameter of a tube or critical distance between two flat plates through which a flame will not propagate. **Flashback** is the phenomenon of propa-

gation of the flame back towards upstream of the burner. This occurs when the flame propagation speed is much higher than the gas exit velocity and the flame eventually travels upstream of the flow. An upstream movement of the flame towards unburned reactants can cause safety issues like explosions in the upstream premixed tank. In such scenarios, the phenomenon of flame extinction through small passageways is used in the design of many practical flame-arresting devices. The upstream travelling flames are thereby quenched and do not pose a safety risk. Quenching distances are determined experimentally using tube burners for quenching diameters, and high aspect ratio slot burners (rectangular) for quenching distances between two parallel flat plates. Williams [22] provides the following two rules of thumb governing ignition and its converse, flame extinction.

- Ignition will occur only if enough energy is added to the gas to heat a slab about as thick as a steady propagating laminar flame to the adiabatic flame temperature.
- Extinction occurs when the rate of liberation of heat by chemical reactions inside the slab approximately balances the rate of heat loss from the slab by thermal conduction.

Simplified Quenching Analysis

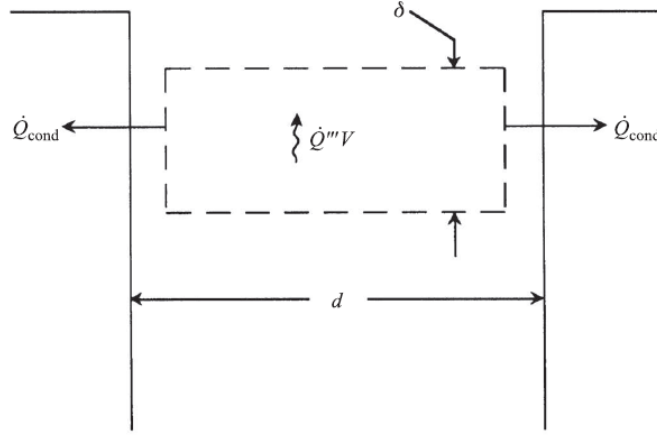


Figure 6: Schematic of flame quenching between two parallel plates.

Energy balance: Equate heat produced by chemical reactions to heat loss by conduction to walls:

$$\dot{Q}'''V = \dot{Q}_{\text{cond,tot}} \quad (31)$$

Volumetric heat release rate $\dot{Q}'''$ is related to $\bar{m}_F'''$,

$$\dot{Q}''' = -\bar{m}_F''' \Delta h_c \quad (32)$$

where

$$\bar{m}_F''' \equiv \frac{1}{(T_b - T_u)} \int_{T_u}^{T_b} \dot{m}_F''' dT \quad (33)$$

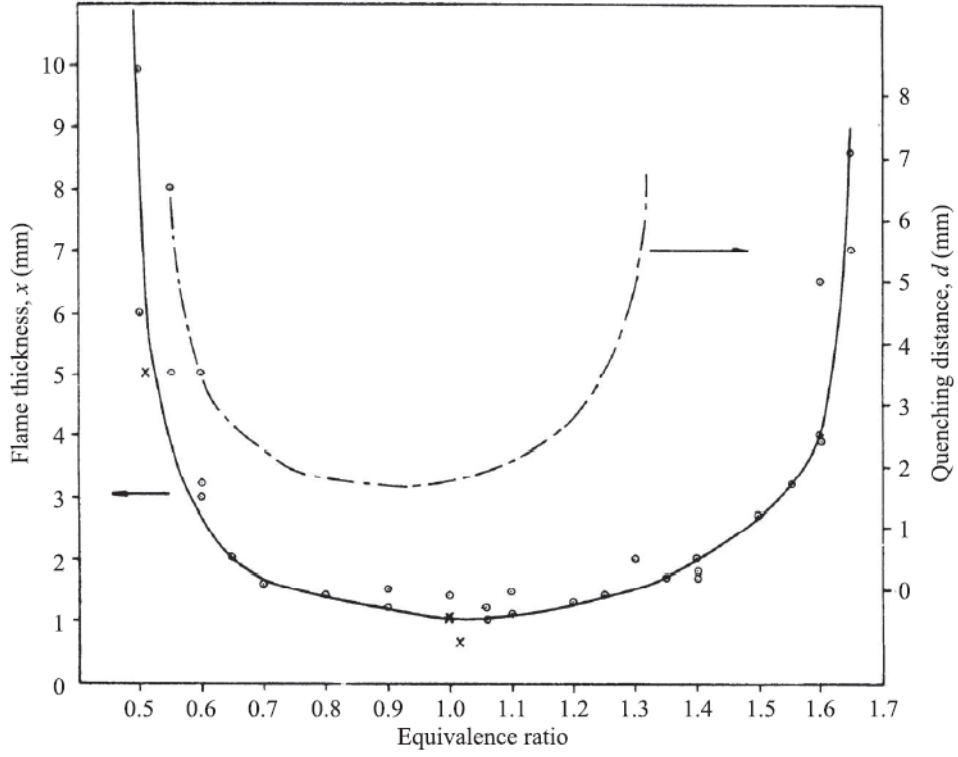


Figure 7: Flame thickness for laminar methane–air flames at atmospheric pressure. Also shown is the quenching distance.

Heat loss by conduction from the slab to the wall: Fourier’s law:

$$\dot{Q}_{\text{cond}} = -kA \left. \frac{dT}{dx} \right|_{T_w} \quad (34)$$

where both temperature gradient, dT/dx , and k are evaluated at gas temperatures at the wall.

$$A = 2\delta L \quad (35)$$

where L is the slot width. The temperature gradient dT/dx is not straightforward to evaluate. A lower bound would be:

$$\frac{dT}{dx} > \frac{(T_b - T_w)}{(d/2)} \quad (36)$$

where a linear distribution from centerline to the wall is assumed. dT/dx is likely to be greater than this, so we introduce a constant b defined as

$$\left| \frac{dT}{dx} \right| \equiv \frac{(T_b - T_w)}{(d/b)} \quad (37)$$

where $b > 2$. Using (32),(34),(37) in quenching criterion, (31), gives

$$(-\bar{m}_F''' \Delta h_c)(\delta d L) = k(2\delta L) \frac{(T_b - T_w)}{(d/b)} \quad (38a)$$

$$d^2 = \frac{2kb(T_b - T_w)}{-\bar{m}_F''' \Delta h_c} \quad (38b)$$

(38b) can be simplified by assuming $T_w = T_u$ and using the following relationships:

$$S_L = \left[-2\alpha(\nu + 1) \frac{\bar{m}_F'''}{\rho_u} \right]^{1/2} \quad (39)$$

$$\Delta h_c = (\nu + 1)c_p(T_b - T_u) \quad (40)$$

Then, (38b) reads

$$d = 2\sqrt{b} \alpha / S_L \quad (41a)$$

$$d = \sqrt{b} \delta \quad (41b)$$

2.2 Flammability Limits

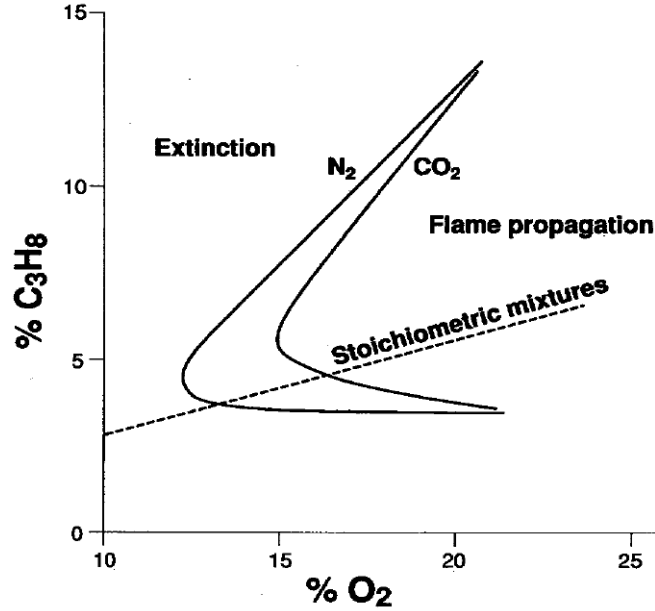


Figure 8: Flammability limits of propane at changing oxygen and propane fractions

A premixed laminar flame will propagate only within a range of mixture strengths. The lower limit is the leanest mixture ($\Phi < 1$) that will allow steady flame propagation, whereas the upper limit represents the richest mixture ($\Phi > 1$). Flammability limits are frequently quoted as percent of fuel by volume in the mixture, or as a percentage of the stoichiometric fuel requirement. Flammability limits are determined experimentally using a "standard" tube method. It is ascertained whether or

not a flame initiated at the bottom of a vertical tube propagates the length of the tube. Flammability limits must be measured under conditions which are not influenced by quenching effects, that is the reaction tube must be of suitably larger diameter. The ignition source must be of sufficient energy to guarantee ignition, otherwise the property under investigation would be that of the limiting ignition energy and not of flammability.

With experimental measurements, it is noted that there are enormous variations between fuels. For example, the flammability range in air for acetylene is 2.5-80% by vol whereas for propane the range is 2.2-9.5%. In some systems the limit seem to correlate with a minimum flame temperature (about 1400 K for methane). The existence of flammability limits is predicted by theories of flame propagation provided the heat loss from the burned gas is included.

2.3 Ignition

We limit our discussion to ignition by electrical sparks and, in particular, focus on the concept of minimum ignition energy. Spark ignition is perhaps the most frequently employed means of ignition in practical devices: for example, spark-ignition and gas-turbine engines, and various industrial, commercial, and residential burners. Spark ignition is highly reliable and does not require a preexisting flame, as required by pilot ignition. In the following, a simple analysis is presented from which the pressure, temperature, and equivalence ratio dependencies of the minimum ignition energy are determined.

Simplified Ignition Analysis We use the second criterion of ignition and quenching applied now to a spherical volume of gas, which represents the incipient propagating flame created by a point spark. Define a critical gas volume radius such that a flame will not propagate if the actual radius is smaller than the critical value. The following step is to assume that the minimum ignition energy to be supplied by the spark is the energy required to heat the critical gas volume from its initial state to the flame temperature.

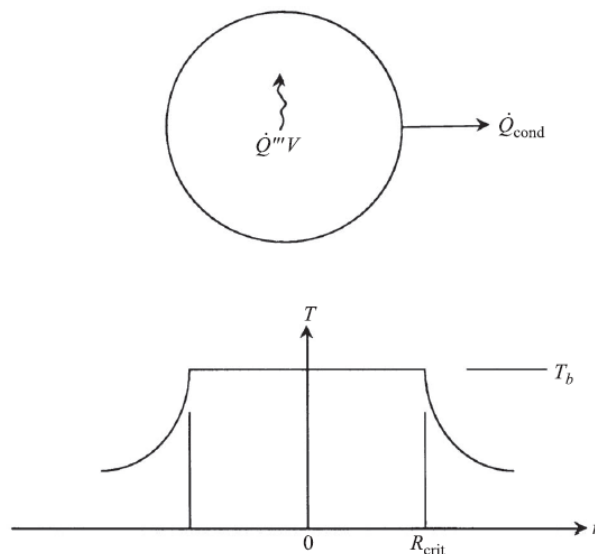


Figure 9: Critical volume of gas for spark ignition

Equate rate of heat liberated by reaction to rate of heat loss to the cold gas by conduction to determine the critical radius:

$$\dot{Q}'''V = \dot{Q}_{\text{cond}} \quad (42)$$

$$\frac{-\bar{m}_F''' \Delta h_c 4\pi R_{\text{crit}}^3}{3} = -k4\pi R_{\text{crit}}^2 \left. \frac{dT}{dr} \right|_{R_{\text{crit}}} \quad (43)$$

$(dT/dr)_{\text{crit}}$ can be obtained from the temperature distribution in $(R_{\text{crit}} \leq r \leq \infty)$

$$\left. \frac{dT}{dr} \right|_{R_{\text{crit}}} = -\frac{(T_b - T_u)}{R_{\text{crit}}} \quad (44)$$

Substitute (44) into (43),

$$R_{\text{crit}}^2 = \frac{3k(T_b - T_u)}{-\bar{m}_F''' \Delta h_c} \quad (45)$$

Using (21), we can express $\bar{m}_F'''$ in terms of S_L or δ . In addition, we have $\Delta h_c = (\nu + 1)c_p(T_b - T_u)$, then (45) becomes

$$R_{\text{crit}} = \sqrt{6} \frac{\alpha}{S_L} \quad (46)$$

where $\alpha = k/(\rho c_p)$. In terms of δ , R_{crit} is,

$$R_{\text{crit}} = (\sqrt{6}/2)\delta \quad (47)$$

It should be noted that $\sqrt{6}/2$ is just an order of magnitude, and should not be taken as a precise constant. Therefore, the critical radius is a few times larger than the flame thickness. Assuming the energy added by the spark heats the critical volume to the burned-gas temperature,

$$E_{\text{ign}} = m_{\text{crit}} c_p (T_b - T_u) \quad (48)$$

where $m_{\text{crit}} = \rho_b 4\pi R_{\text{crit}}^3/3$, then

$$E_{\text{ign}} = 61.6 \rho_b c_p (T_b - T_u) (\alpha/S_L)^3 \quad (49)$$

Eliminating ρ_b using ideal gas law,

$$E_{\text{ign}} = 61.6 P \left(\frac{c_p}{R_b} \right) \left(\frac{T_b - T_u}{T_b} \right) \left(\frac{\alpha}{S_L} \right)^3 \quad (50)$$

Pressure and Temperature Dependency From (50), we can assess the P dependency of ignition energy. We know from previous analysis that

$$\alpha \propto T_u \bar{T}^{0.75} P^{-1} \quad (51)$$

From (25) (with $n \approx 2$)

$$S_L \propto \bar{T}^{0.375} T_u T_b^{-\frac{n}{2}} \exp \left[\frac{-E_A}{2R_u T_b} \right] P^{\frac{n-2}{2}} \quad (52)$$

The combined effects from S_L and α yield

$$E_{\text{ign}} \propto P^{-2} \quad (53a)$$

which agrees extremely well with the experimental results. As T_u is increased the minimum ignition energy decreases.

$$E_{\text{ign}} \propto T_u^{-x} \quad (53b)$$

3 Flame Stabilisation

Important design criteria for flame stabilisation includes avoiding flashback and liftoff. In flashback, flame enters and propagates through the burner upstream without quenching. In liftoff, flame is not attached to the burner, but stabilized at a distance from it. Flashback is a safety hazard and liftoff is correlated to issues of incomplete burning, ignition problems, and control of the flame position.

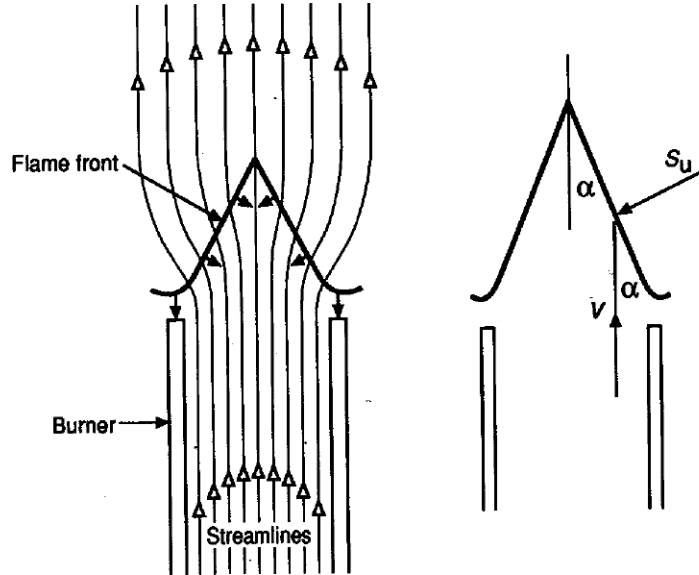


Figure 10: Stabilisation of premixed flame front at the locus of points where laminar flame speed equals normal component of unburned gas velocity

Both flashback and liftoff are related to matching the local laminar flame speed to the local flow velocity. A flame can be stabilized on the burner only between certain flow velocity limits. If the gas velocity is progressively reduced, a point will be reached eventually at which the burning velocity exceeds the gas velocity somewhere across the burner. At this point, flame will propagate back down the burner. On the basis of increasing flow velocity, the flame structure can be categorised as shown in the Fig. ??

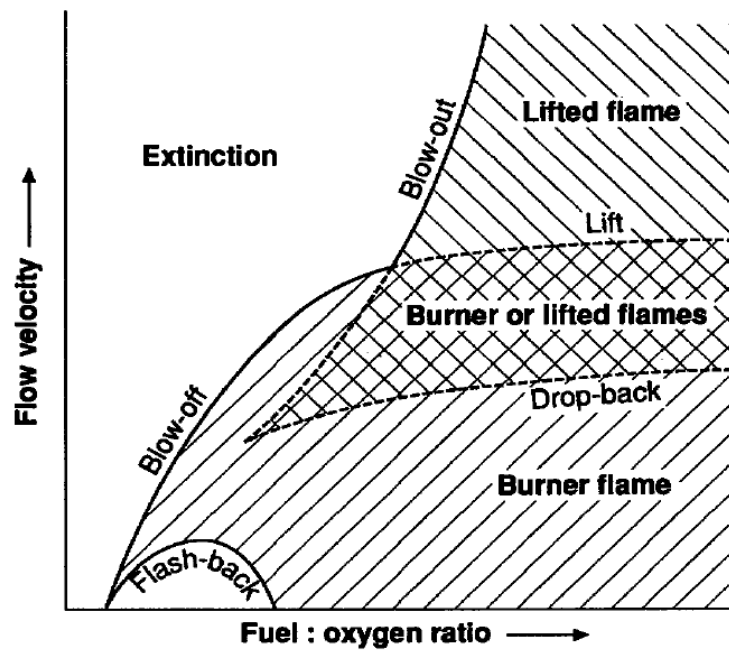


Figure 11: Stability diagrams for flashback, liftoff, extinction, and blow off.